POKHARA UNIVERSITY

Level: Bachelor	Semester: Fall	Year : 2024
Programme: BE		Full Marks : 100
Course: Calculus II		Pass Marks : 45
		Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Evaluate: $\int_0^{\pi}\int_x^{\pi}\frac{\sin y}{y}\,dydx$ by changing the order of integration. 5

   b) Evaluate the integrals: $\iiint_V (x^2+y^2+z^2)dxdydz$ taken over the volume of the sphere $x^2+y^2+z^2=1$. 5

   c) Find the volume of the solid whose base is the region in the xy-plane that is bounded by the parabola $y=4-x^2$ and the line $y=3x$, while the top of the solid is bounded by the plane $z=x+4$. 5

2. a) Solve the differential equation: $y''+9y=0$ by using power series method. 7

   b) Define Legendre differential equation. Also, derive the solution of Legendre differential equation. 8

OR

If $J_n(x)$ represents the Bessel's function of order n, then show that:

i. $\frac{d}{dx}[x^n J_n(x)] = x^n J_{n-1}(x)$ 4

ii. $J_{\frac{1}{2}}(x)=\sqrt{\frac{2}{\pi x}}\sin x$ 4

3. a) Solve the following differential equation by using Laplace transform $y''+2y'-3y=6e^{-2t}, y(0)=2, y'(0)=-14$. 7

   b) i) State second shifting property for Laplace transform and using this, find the Laplace transform of $e^{-3t}u(t-3)$. 4

   ii) Find the inverse Laplace transform of $\frac{\pi}{(s-3)^2+\pi^2}$. 4

4. a) Find the directional derivative of $f=x^2yz+4xz^2$ at $(1,-2,1)$ in the direction $2\vec{i}-\vec{j}-2\vec{k}$. 5

   b) Show that the vector $\vec{F}=(x^2-yz,\ y^2-xz,\ z^2-xy)$ is conservative vector field and find the scalar potential function $\emptyset$ such that $\vec{F}=\nabla\emptyset$. 5

   c) Find the work done by the force $\vec{F}=(3x^2,\ 2xz-y,\ z)$ along the curve $x=t,\ y=\frac{t^2}{4},\ z=\frac{3t^2}{8}$ from (0, 0, 0) to (2, 1, 3). 5

5. a) Evaluate the integral by using Green's theorem 7

   $\oint_C [(3x^2+y)\,dx+4y^2\,dy]$ where C: the boundary of the triangle with vertices (0, 0), (1, 0), (0, 2); oriented in counterclockwise direction.

   b) State Stoke's Theorem. Evaluate $\oint_C(\vec{F}.d\vec{r})$ by using Stoke's theorem, where $\vec{F}=(y,\ xz^3,-zy^3)$ and $C: x^2+y^2=4,\ z=3$. 8

OR

Evaluate: $\iint_S \vec{F}.\vec{n}\,dA$, where $\vec{F}=(18z,\ -12,\ 3y)$ and S is the surface of plane $2x+3y+6z=12$ in the first octant.

6. a) Find the Fourier series of $f(x)=x+|x|,\ -\pi<x<\pi$ and show that $\frac{\pi^2}{8}=1+\frac{1}{9}+\frac{1}{25}+\cdots$. 7

   b) Find the Fourier sine and cosine series for the function $f(x)=x^2$ in the interval $0<x<L$. 8

7. Write short notes on: **(Any two)** 2×5

   a) Find the general solution of the partial differential equation $u_x+u_y=u$.

   b) Derive the one-dimensional traffic flow model using conservation law.

   c) A particle moves along the curve $x=t^3+1,\ y=t^2,\ z=2t+5$. Find the velocity and acceleration at $t=1$.